

## LECTURE 1

# From the Naturals to the Reals: Induction, Ordered Fields, and Completeness

MATH S4061 | Introduction to Modern Analysis I

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*God made the integers; all else is the work of man.*

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**Overview.** The course opens with a deceptively simple question—is  $0.999\dots = 1$ ?—whose honest answer requires building the real numbers from the ground up. We begin with the natural numbers and induction, construct  $\mathbb{Z}$  and  $\mathbb{Q}$ , and isolate the field and order axioms they satisfy. The rationals then fall short for analysis:  $\sqrt{2}$  is missing, and the sets bounding it have neither a largest nor a smallest element. Repairing that gap is the *least-upper-bound property*, which characterises  $\mathbb{R}$ ; from it follow the Archimedean property and the density of  $\mathbb{Q}$ —and, at last, the resolution of  $0.999\dots = 1$ . The companion text is Rudin, Chapter 1, especially the ordered-field and least-upper-bound material [R 1.5](#)–[R 1.20](#).

# 1.1 The Natural Numbers and Induction

## Definition 1.1.1

The natural numbers

The *natural numbers* are  $\mathbb{N} = \{0, 1, 2, 3, \dots\}$ <sup>1</sup>, generated from 0 by the *successor function*

$$S : \mathbb{N} \rightarrow \mathbb{N}, \quad n \mapsto n + 1.$$

## Axiom 1.1.2

Induction

If  $A \subseteq \mathbb{N}$  satisfies  $0 \in A$  and  $n \in A \Rightarrow n + 1 \in A$ , then  $A = \mathbb{N}$ . Rudin treats induction as prerequisite proof background rather than a numbered analysis result.

## Theorem 1.1.3

The principle of mathematical induction

Let  $P(n)$  be a statement for each  $n \in \mathbb{N}$ . If  $P(0)$  holds, and  $P(n) \Rightarrow P(n + 1)$  for every  $n$ , then  $P(n)$  holds for all  $n \in \mathbb{N}$ . This is the proof principle behind later uses of induction; Rudin assumes it as background.

## Proof 1.1.3

*Proof of the Induction Principle from the Axiom.*

- (1) Let  $A = \{n \in \mathbb{N} : P(n) \text{ holds}\}$ .
- (2) Since  $P(0)$  holds,  $0 \in A$ ; and  $P(n) \Rightarrow P(n + 1)$  says  $n \in A \Rightarrow n + 1 \in A$ .
- (3) By the induction axiom (1.1.2),  $A = \mathbb{N}$ ; that is,  $P(n)$  holds for every  $n$ .

■

[Direct] uses 1.1.2.

## Example 1.1.4

*Gauss's sum*

For every integer  $n \geq 1$ ,  $1 + 2 + \dots + n = \frac{n(n + 1)}{2}$ . This is included as induction practice rather than as a Rudin-numbered item.

## Proof 1.1.4

*Proof by Induction of Gauss's Formula.*

- (1) For  $n = 1$  the formula reads  $1 = \frac{1 \cdot 2}{2}$ , which holds.

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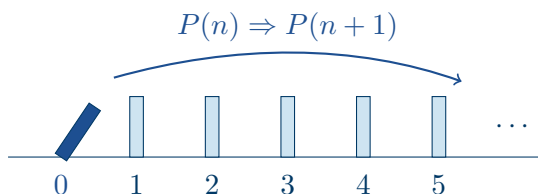
<sup>1</sup>We take  $0 \in \mathbb{N}$ ; Rudin instead starts the positive integers at 1. The convention is harmless—only the base of an induction shifts (Gauss's sum, 1.1.4, begins at  $n = 1$ ).

(2) Assume  $1 + \cdots + n = \frac{n(n+1)}{2}$ . Adding  $n+1$  to both sides,

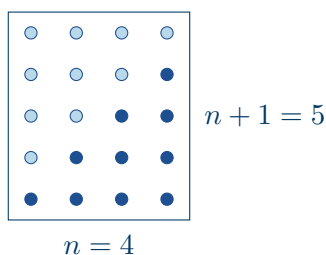
$$1 + \cdots + n + (n+1) = \frac{n(n+1)}{2} + (n+1) = \frac{(n+1)(n+2)}{2},$$

which is the formula for  $n+1$ . The result follows by induction (1.1.3). ■

[A10] from the base  $n = 1$  (apply 1.1.3 to the shifted statement).



**Figure 1.1.5.** INDUCTION AS TOPPLING DOMINOES. The base case  $P(0)$  topples the first domino; the inductive step  $P(n) \Rightarrow P(n+1)$  passes the fall to the next, so every  $P(n)$  falls in turn—the conclusion of 1.1.2 and 1.1.3.



**Figure 1.1.6.** GAUSS'S SUM AS HALF A RECTANGLE. Two copies of the staircase  $1 + 2 + 3 + 4$  interlock into a  $4 \times 5$  grid, so  $2(1 + 2 + 3 + 4) = 4 \cdot 5$ ; in general  $1 + \cdots + n = \frac{n(n+1)}{2}$  (1.1.4).

## 1.2 The Integers and the Rationals

### Construction 1.2.1

#### The integers

The *integers* are signed naturals with the two zeros glued together:

$$\mathbb{Z} := (\{+, -\} \times \mathbb{N}) / \sim, \quad (+, 0) \sim (-, 0).$$

Writing  $+n, -n$  for the classes recovers  $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ . Rudin treats  $\mathbb{Z}$  as background and begins the analysis notation with the rational field in R 1.4.

**Construction 1.2.2****The rationals**

The *rationals* are equivalence classes of integer pairs with nonzero second coordinate:

$$\mathbb{Q} := \{(p, q) \in \mathbb{Z} \times \mathbb{Z} : q \neq 0\} / \sim, \quad (p, q) \sim (r, s) \iff ps = rq.$$

The class of  $(p, q)$  is written  $\frac{p}{q}$ . Rudin records the rational-number notation and convention in R 1.4.

**Remark 1.2.3**

*One number, many fractions*

A rational *is* the whole class, e.g.  $[\frac{1}{2}] = \{\frac{1}{2}, \frac{2}{4}, \frac{-1}{-2}, \frac{17}{34}, \dots\}$ . In practice one names it by its *preferred representative*: the irreducible fraction  $\frac{a}{b}$  with  $\gcd(a, b) = 1$  and the sign carried in the numerator. This is the equivalence-class bookkeeping behind the notation in R 1.4.

## 1.3 Fields and Ordered Fields

**Definition 1.3.1****Field**

A *field* is a set  $F$  with two operations  $+: F \times F \rightarrow F$  and  $\cdot: F \times F \rightarrow F$  satisfying eleven axioms<sup>2</sup>—making  $F$  an abelian group under  $+$ , making  $F \setminus \{0\}$  an abelian group under  $\cdot$ , with multiplication distributing over addition.

**Definition 1.3.2****Order**

An *order* on a set  $S$  is a relation  $<$  such that (cf. R 1.5)

- (1) (*trichotomy*) for  $x, y \in S$ , exactly one of  $x < y$ ,  $y < x$ ,  $x = y$  holds;
- (2) (*transitivity*)  $x < y$  and  $y < z$  imply  $x < z$ .

A set carrying an order is *totally* (or *linearly*) ordered. Rudin calls this an ordered set in R 1.6.

**Definition 1.3.3****Ordered field**

An *ordered field* is a field  $F$  with an order  $<$  compatible with the operations (R 1.17):

- (1)  $y < z \Rightarrow x + y < x + z$  for all  $x$ ;
- (2)  $x > 0$  and  $y > 0 \Rightarrow xy > 0$ .

**Remark 1.3.4**

*The rationals are an ordered field*

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<sup>2</sup>Five for addition (A1–A5), five for multiplication (M1–M5), and the distributive law (D); the full statements are R 1.12.

$\mathbb{Q}$  is an ordered field. The rest of the lecture asks what it *lacks*—and why that makes it unfit for analysis. Compare Rudin’s ordered-field definition R 1.17 and the real-field existence theorem R 1.19.

## 1.4 Why the Rationals Are Not Enough

### Proposition 1.4.1

No rational squares to 2

There is no  $x \in \mathbb{Q}$  with  $x^2 = 2$  (R 1.1).

#### Proof 1.4.1

*Proof by Contradiction of the Irrationality of  $\sqrt{2}$ .*

- (1) Suppose  $x = \frac{p}{q} \in \mathbb{Q}$  is irreducible with  $x^2 = 2$ . Then  $p^2 = 2q^2$ , so  $p^2$  is even, hence  $p$  is even (an odd integer has an odd square); write  $p = 2k$ .
- (2) Substituting,  $4k^2 = 2q^2$ , so  $q^2 = 2k^2$  is even, hence  $q$  is even.
- (3) Then  $2 \mid \gcd(p, q)$ , contradicting that  $\frac{p}{q}$  is irreducible. So no such  $x$  exists. ■

[A9] cf. R 1.1; uses “ $n^2$  even  $\Rightarrow n$  even”.

### Proposition 1.4.2

The rationals have a gap

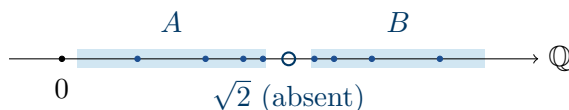
Let  $A = \{p \in \mathbb{Q} : p > 0, p^2 < 2\}$  and  $B = \{p \in \mathbb{Q} : p > 0, p^2 > 2\}$ . Then  $A$  has no largest element and  $B$  has no smallest element.

#### Proof 1.4.2

*Proof by an Explicit Construction of the Gap.*

- (1) Given  $p > 0$ , set
 
$$q = p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2} > 0.$$
- (2) A computation gives
 
$$q^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2},$$
 so  $q^2 - 2$  has the same sign as  $p^2 - 2$ : if  $p \in A$  then  $q \in A$ , and if  $p \in B$  then  $q \in B$ .
- (3) Also  $q - p = -\frac{p^2 - 2}{p + 2}$ , so  $q > p$  when  $p \in A$  and  $q < p$  when  $p \in B$ .
- (4) Hence every  $p \in A$  is exceeded by a larger  $q \in A$ , so  $A$  has no largest element; and every  $p \in B$  is undercut by a smaller  $q \in B$ , so  $B$  has no smallest element. ■

[Constr] cf. R 1.1.



**Figure 1.4.3.** THE GAP AT  $\sqrt{2}$ . The positive rationals  $A$  ( $p^2 < 2$ ) climb toward  $\sqrt{2}$  with no largest member;  $B$  ( $p^2 > 2$ ) descend toward it with no smallest; in  $\mathbb{Q}$  the meeting point  $\sqrt{2}$  is simply absent.

## 1.5 Bounds and the Supremum

### Definition 1.5.1

#### Upper bound

Let  $S$  be an ordered set and  $E \subseteq S$ . An element  $\beta \in S$  is an *upper bound* for  $E$  if  $x \leq \beta$  for every  $x \in E$  (R 1.7).

### Definition 1.5.2

#### Least upper bound (supremum)

$\beta$  is the *least upper bound* of  $E$  if it is an upper bound for  $E$  and no smaller element is:  $\alpha < \beta$  implies  $\alpha$  is not an upper bound. We write  $\beta = \sup E = \text{lub } E$  (R 1.8).

### Definition 1.5.3

#### Lower bound and infimum

Dually,  $\beta \in S$  is a *lower bound* for  $E$  if  $\beta \leq x$  for every  $x \in E$ , and the *greatest lower bound* (or *infimum*) if no larger element is a lower bound:  $\beta < \alpha$  implies  $\alpha$  is not a lower bound. We write  $\beta = \inf E = \text{glb } E$  (cf. R 1.8). Since  $\inf E = -\sup(-E)$ , the least-upper-bound property (1.6.1) carries a matching greatest-lower-bound property.

## 1.6 The Real Number System

### Theorem 1.6.1

#### The real number system

There is an ordered field  $\mathbb{R} \supseteq \mathbb{Q}$  in which every nonempty subset that is bounded above has a least upper bound. It is unique up to isomorphism.<sup>3</sup> This is Rudin’s existence theorem for the real field (R 1.19).

### Remark 1.6.2

#### Completeness and density

The property in 1.6.1 is *completeness*:  $\mathbb{R}$  is complete iff it has the least-upper-bound property. A first consequence is that  $\mathbb{Q}$  is *dense* in  $\mathbb{R}$ —between any two reals lies a rational (1.7.2; cf. R 1.19–R 1.20).

<sup>3</sup>An isomorphism of ordered fields is a bijection preserving  $+$ ,  $\cdot$ , and  $<$ ; “unique up to isomorphism” means any two complete ordered fields are indistinguishable as such. Two were built in 1872: Dedekind’s cuts (*Stetigkeit und irrationale Zahlen*) and Cantor’s equivalence classes of Cauchy sequences of rationals.

**Remark 1.6.3**

*Three ways to build  $\mathbb{R}$*

- *Accept it*—take 1.6.1 on faith and learn to use sup.
- *Dedekind cuts* (1.6.4)—realise each real as a downward-closed set of rationals.
- *Completion*—the general procedure that completes any metric space (deferred).

Rudin states the existence theorem abstractly in R 1.19; these are model-building viewpoints.

**Definition 1.6.4****Dedekind cut**

A *cut* is a set  $\alpha \subseteq \mathbb{Q}$  such that

- (1)  $\alpha \neq \emptyset$  and  $\alpha \neq \mathbb{Q}$ ;
- (2)  $\alpha$  is downward closed:  $r \in \alpha$  and  $s < r$  imply  $s \in \alpha$ ;
- (3)  $\alpha$  has no largest element.

The cuts, suitably ordered, *are* the real numbers; this is one construction behind Rudin's real-field theorem R 1.19.

**Example 1.6.5**

$\sqrt{2}$  as a cut

The real number  $\sqrt{2}$  is the cut

$$\alpha = \{q \in \mathbb{Q} : q < 0\} \cup \{q \in \mathbb{Q} : q^2 < 2\}.$$

This is the set  $A$  of 1.4.2 together with the non-positive rationals; the former “gap” is now a bona fide element (cf. R 1.1, R 1.19).

## 1.7 The Archimedean Property, Density, and Decimals

**Theorem 1.7.1****Archimedean property**

If  $x, y \in \mathbb{R}$  and  $x > 0$ , then there is a positive integer  $n$  with  $nx > y$ .<sup>4</sup> This is the Archimedean part of R 1.20.

**Proof 1.7.1**

*Proof by Contradiction of the Archimedean Property.*

- |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <ol style="list-style-type: none"> <li>(1) Suppose not: the set <math>A = \{nx : n \in \mathbb{N}\}</math> is bounded above. By the least-upper-bound property (1.6.1) it has a supremum <math>\alpha = \sup A</math>.</li> <li>(2) Since <math>x &gt; 0</math>, <math>\alpha - x &lt; \alpha</math>, so <math>\alpha - x</math> is not an upper bound of <math>A</math>: some <math>m \in \mathbb{N}</math> has <math>mx &gt; \alpha - x</math>.</li> </ol> |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

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<sup>4</sup>Named by Otto Stolz in the 1880s after Archimedes, who postulated it in *On the Sphere and the Cylinder*; the idea is older still—Euclid records it (*Elements* V, Def. 4) and credits Eudoxus.

- (3) Then  $(m+1)x > \alpha$ , yet  $(m+1)x \in A$ —contradicting  $\alpha = \sup A$ . So  $A$  is not bounded above, and in particular some  $nx > y$ . ■

[A5+A9] cf. R 1.20; via the supremum; uses 1.6.1.

### Theorem 1.7.2

#### Density of $\mathbb{Q}$ in $\mathbb{R}$

If  $x, y \in \mathbb{R}$  and  $x < y$ , then there is a rational  $q$  with  $x < q < y$ . This is the density part of R 1.20.

#### Proof 1.7.2

*Proof by the Archimedean Property of the Density of  $\mathbb{Q}$ .*

- (1) Since  $y - x > 0$ , the Archimedean property (1.7.1) gives  $n \in \mathbb{N}$  with  $n(y - x) > 1$ , i.e.  $ny - nx > 1$ .  
 (2) Since  $ny - nx > 1$ , the interval  $(nx, ny)$  contains an integer  $m$  (§1.A.1).  
 (3) Dividing by  $n > 0$ ,  $x < \frac{m}{n} < y$ , so  $q = \frac{m}{n}$  works. ■

[Direct] cf. R 1.20; uses 1.7.1.

### Construction 1.7.3

#### Real numbers as decimals

Every *nonnegative* real is the supremum of its finite decimal truncations (a negative real is then named by its sign):

$$1.4112\dots \mapsto \sup\{1, 1.4, 1.41, 1.411, \dots\}.$$

This uses the least-upper-bound and Archimedean machinery in R 1.19–R 1.20.

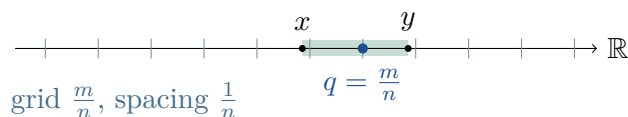
### Example 1.7.4

$$0.999\dots = 1$$

This settles the opening question. Put  $x = 1 - 0.999\dots$ ; then  $0 \leq x < 10^{-k}$  for every  $k \in \mathbb{N}$ . By the Archimedean property (1.7.1)<sup>5</sup> no positive real lies below every  $10^{-k}$ , so  $x = 0$ —that is,  $0.999\dots = 1$ . More generally every terminating decimal equals its repeating-9 twin ( $2.5 = 2.4999\dots$ , and so on), so

$$\mathbb{R} = \{\text{decimal expansions}\} / \sim,$$

the quotient that identifies each terminating decimal with that twin (cf. R 1.20).



<sup>5</sup>If  $x > 0$ , the Archimedean property gives  $n$  with  $nx > 1$ ; taking  $k$  with  $10^k \geq n$  then yields  $10^k x > 1$ , i.e.  $x > 10^{-k}$ .



**Figure 1.7.5.** DENSITY OF  $\mathbb{Q}$  IN  $\mathbb{R}$ . Choosing  $n$  with  $1/n < y - x$  makes the grid of multiples  $m/n$  finer than the gap, so some  $m/n$  lands strictly between  $x$  and  $y$  (1.7.2).

# Appendix

*Supplementary material—additional proofs, context, and other treatments; not part of the lecture proper.*

## Supplement 1.A.1

*An interval longer than 1 contains an integer*

If  $s, t \in \mathbb{R}$  with  $t - s > 1$ , then there is an integer  $m$  with  $s < m < t$ . (Used silently in the density proof, 1.7.2.)

### Proof 1.A.1

*Direct Proof from the Archimedean Property.*

- (1) The integers exceeding  $s$  are nonempty (by the Archimedean property) and bounded below by  $s$ , so they have a least element  $m$ ; thus  $m - 1 \leq s < m$ .
- (2) Hence  $m \leq s + 1 < t$ , and therefore  $s < m < t$ .

■

[Direct] uses 1.7.1; cf. R 1.20.

## Supplement 1.A.2

*Every nonnegative real has a decimal expansion*

Construction 1.7.3 names each nonnegative real as the supremum of its truncations; here we build those truncations digit by digit and confirm the supremum is the number we began with. Given  $x \geq 0$ , there are an integer  $a_0 \geq 0$  and digits  $d_1, d_2, \dots \in \{0, \dots, 9\}$  whose truncations

$$t_k = a_0 + \sum_{i=1}^k \frac{d_i}{10^i} \quad \text{satisfy} \quad 0 \leq x - t_k < 10^{-k},$$

so that  $x = \sup_k t_k$ . The rule below always takes the *largest* admissible digit, so a number with a terminating expansion receives it ( $1 = 1.000\dots$ ), and the repeating-9 twin (1.7.4) is the single expansion this rule never produces. The key order input is R 1.20.

### Proof 1.A.2

*Direct Proof by Greedy Digit Selection.*

- (1) For the integer part, let  $a_0$  be the greatest integer with  $a_0 \leq x$  — it exists by the least-integer argument of 1.A.1, applied to the integers exceeding  $x$ , and  $a_0 \geq 0$  since  $x \geq 0$ . Then  $t_0 = a_0$  obeys  $0 \leq x - t_0 < 1$ .
- (2) For the digits, suppose  $t_{k-1}$  has been built with  $0 \leq x - t_{k-1} < 10^{-(k-1)}$ . Let  $d_k$  be the largest digit in  $\{0, \dots, 9\}$  with  $t_{k-1} + d_k 10^{-k} \leq x$ ; the digit 0 qualifies, and  $d_k \leq 9$  because  $x < t_{k-1} + 10 \cdot 10^{-k} = t_{k-1} + 10^{-(k-1)}$ . Put  $t_k = t_{k-1} + d_k 10^{-k}$ . Then  $t_k \leq x$ , while maximality of  $d_k$  gives  $x < t_k + 10^{-k}$ ; hence  $0 \leq x - t_k < 10^{-k}$  and the invariant persists.
- (3) For the supremum: every  $t_k \leq x$ , so  $x$  is an upper bound. If  $\beta < x$ , then  $x - \beta > 10^{-k}$  for some  $k$  (Archimedean property, 1.7.1), so  $t_k > x - 10^{-k} > \beta$ ; thus no

$\beta < x$  bounds the truncations, and  $x = \sup_k t_k$ .  
 [Direct] uses 1.7.1, 1.A.1.

■

# Index of Objects

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